

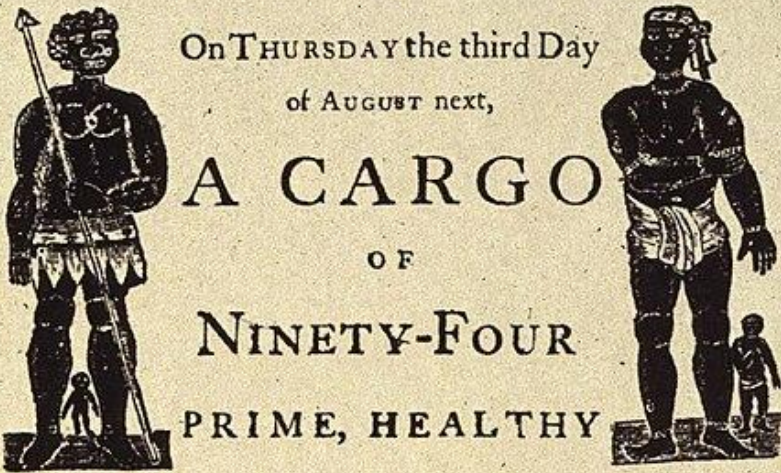
Charting Human Cargo: Digital Analysis of the Transatlantic Slave Trade through Slave Voyages

Charlestown, July 24th, 1769.

TO BE SOLD,

On THURSDAY the third Day
of AUGUST next,

A CARGO
OF
NINETY-FOUR
PRIME, HEALTHY



NEGROES,

CONSISTING OF
Thirty-nine MEN, Fifteen BOYS,
Twenty-four WOMEN, and
Sixteen GIRLS.

JUST ARRIVED,
In the Brigantine DEMBIA, *Francis Bare*, Master, from SIERRA-
LEON, by
DAVID & JOHN DEAS.

Reproduction of a handbill advertising a slave auction in Charleston, British Province of South Carolina, in 1769

GitHub link for the exam project: [Here!](#)

Introduction

“We must either become slaves or fly to arms. I shall not (and hope no Americans will) hesitate one moment which to choose.” Samuel Ward, a delegate from Rhode Island wrote to John Dickinson from Pennsylvania in 1774. He wrote that “Even death itself was infinitely preferable to slavery, which in one word comprehends poverty, misery, infamy and every species of ruin and destruction.”¹

The transatlantic slave trade involved the transportation of scores of enslaved people across the Atlantic by ship. The enslaved people were from Africa who were taken from their home and across the seas to the Americas. This trade was operated by slave ships from both Europe and the New World. The Europeans established a coastal slave trade in the 15th century and began exporting the enslaved to the Americas in the 16th century through to the 19th.² The vast majority of those who were taken were from western and central Africa. The major Atlantic slave trading nations, in order of trade volume, were Portugal, Britain, Spain, France, the Netherlands, the United States, and Denmark. Several had established outposts on the African coast, where they purchased slaves from local African leaders.³ The enslaved were malnourished when they were brought aboard the ships, brought on by the long trip to the coast. The English and Portuguese governors at the coast gave them only the minimum to survive. The grim reality of the situations deepens when we learn of the their living situations on the ships themselves as they were forced into confined and overcrowded quarters where they would be chained and made to wait as their only sustenance came in rice and yams who were prone to rot and going rancid. As such the enslaved would live in a prolonged state of malnutrition making them vulnerable to gastrointestinal diseases. This, among many other reasons, oft led to the enslaved perishing on the long voyage before they arrived at the Americas.⁴

Cataloguing and charting the insurmountable amounts of data about these myriad voyages and the struggles of the enslaved was the goal when several prominent historians set out to create SlaveVoyages.org. Historians such as Herbert S. Klein, David Richardson, Stephen Behrendt and David Eltis. Eltis’ own work is among the sources used in this project. A

¹ McBurney, Christian M. “The First Efforts to Limit the African Slave Trade Arise in the American Revolution Part 3 of 3: Congress Bans the African Slave Trade.” *Journal of the American Revolution*, September 23, 2020.

² Diffie, Bailey (1963). *Prelude to Empire: Portugal Overseas Before Henry the Navigator*. University of Nebraska Press. p. 58.

³ Klein, Herbert S.; Klein, Jacob (1999). *The Atlantic Slave Trade*. Cambridge University Press. pp. 103–139.

⁴ Negrón, Ramona, Jessica Den Oudsten, Camilla de Koning, and Karwan Fatah-Black, eds. *The Dutch Transatlantic Slave Trade: New Methods, Perspectives and Sources*. Leiden: Leiden University Press, 2024. P. 151

historical database is, in its essence, a collection of data about a limited specific time period or historical event. For the vast majority of time these were analog archives one had to visit in person to peruse. When the first digital databases came around, they were hierarchical, only allowing you to access subtopics from larger categories. This was until the introduction of the relational model in the 1970s. Under the relational model data can be accessed from multiple paths and can be sorted from multiple factors.

During this development in the 1970s the first work on what would become SlaveVoyages.org began. The aforementioned scholars began creating datasets of slave voyages. The database itself was formally created in 1999. At this time it was not open source but available on a cd-rom one had to pay for. The first open access web version came in 2008 and it has been steadily updated ever since, the latest update being in 2025. The archive draws on the national archives of twenty-seven countries in Europe, the Americas and Africa.⁵ Today it is an incredibly powerful tool available to historians who wish to research the vast breadth of the transatlantic slave trade and its long-lasting consequences into the modern world.⁶

Thus, we arrive to the research question we will be addressing: **How can the SlaveVoyages digital archive be used to analyze geographical and temporal patterns in the transatlantic slave trade between 1750 and 1800?**

Historical & Archival Context

On SlaveVoyages.org's own methodology page there is an important note to keep in mind when we proceed to use this database. The dataset contains thousands of names of ships and their respective owners but almost no names of the millions of slaves taken from their homes, transported halfway across the world and subsequently forced into a life of labour and shunned to an early grave.⁷ As historian Jessica Marie Johnson discusses in her work 'Markup Bodies' there is no neutrality to be found when discussing the exploitation and enslavement of innocents. Technology has not made the realities of bondage any more palatable or easier to discuss.⁸ When we address the datasets these nameless millions are subsequently treated as numbers. These numbers are not neutral. By treating enslaved people as mere numbers and as cargo we are repeating the violence of the slavers themselves. In recent years, SlaveVoyages.org has expanded its catalogue to now include

⁵ O'Malley, Gregory E., and Alex Borucki. "Creation of the Intra-American Slave Trade Database." *Journal of Slavery and Data Preservation*. P. 5

⁶ O'Malley, Gregory E., and Alex Borucki. "Creation of the Intra-American Slave Trade Database." *Journal of Slavery and Data Preservation*. P. 2

⁷ O'Malley, Gregory E., and Alex Borucki. "Creation of the Intra-American Slave Trade Database." *Journal of Slavery and Data Preservation*. P. 5

⁸ Johnson, Jessica Marie. "Markup Bodies: Black [Life] Studies and Slavery [Death] Studies at the Digital Crossroads." *Social Text* 36, no. 4 (December 1, 2018): page 60

extended research on the African families affected by the transatlantic slave trade. This is a step in the direction of acknowledging the enslaved as more than numbers, as people.

SlaveVoyages.org being a digitized archive allows nigh anyone from anywhere to look into the tragedies of the transatlantic slave trade. As Helle Strandgaard Jensen argues, the increasing availability of sources in digital archives has fostered the need to understand how the logic of these archives influence historical research.⁹ Modern French philosophy of the 1960s and 1970s produced two insights that highlighted this point. The first is that archives are powerful research infrastructures that have influence on both the representation and the reproduction of historiographical traditions. The latter is that historians need critical awareness of the organization of archives and their knowledge if they want to work innovatively with their collections. Those who have raised criticism against traditional archives for reenforcing existing power structures that serve nationalistic, male-oriented political history need knowledge of archival policies as a fundamental part of deconstructing the logic of the archive. As such archival expertise is key in the historian's role of interpreting the past.¹⁰

One of the greatest strengths of SlaveVoyges.org is how it takes fragmented historical records into one cohesive searchable dataset. It gathers data from myriad different historical sources and collects them into cohesive factors, thereafter it allows you to narrow your focus on what you wish to study in particular. However despite its scale and complexity we cannot mistake SlaveVoyages.org as a completed picture of the transatlantic slave trade. As said the database is constructed from many many sources, some of which are incomplete, inconsistent or absent. As a result many voyages are missing, some numbers are estimated and regional classifications simplify historical realities that are far more complex. The clean structure of the dataset can create an illusion of certainty and completeness that we must not accept as a sure fact.

SlaveVoyages demonstrates both the power and the limitations of digital historical archives. As the archive makes large scale quantitative analysis possible through structured and accessible data, at the same time it highlights the simplifications and interpretive choices that are inherent in all archives. The process of transforming fragmented historical data into one collective usable dataset inevitably affects the kinds of questions that can be asked or answered through it.

Methodology

⁹ Helle Strandgaard Nielsen and Niels Brügger, "Digital Archival Literacy for Historians," Centre for Digital History, Aarhus University, accessed May 27, 2026, page 1

¹⁰ Helle Strandgaard Nielsen and Niels Brügger, "Digital Archival Literacy for Historians," Centre for Digital History, Aarhus University, accessed May 27, 2026, page 3

Having chosen to focus on the transatlantic slave trade and the collected data on slavevoyages.org, next came the choice of what data to focus on. The original focus of this project was on the mortality rates of the enslaved aboard the ships during the voyage.

In the end I chose to focus on the five parameters that are: voyage year, origin region, destination region, embarked slaves and disembarked slaves. The mortality rate itself was something I would need to calculate with the help of Rstudio through mutating the numbers from embarked and disembarked slaves. This was done with a focus on the years 1750-1800.

Slavevoyages.org allows you to choose what data you wish to be present, and which should be excluded from the CSV file you download. As such I put in the parameters and soon had a customized CSV named 'trans-atlantic'.

Then opening a new project on Rstudio with my projectfolder as the directory I proceeded with the coding. Using the library(readr) function I loaded the trans-atlantic csv in. The next step was then to convert the raw csv into a more usable and cleaner dataset which was aptly named 'slave_data'. Next came the matter of calculating mortality as well as the most common origin and destination.

Calculating mortality was a matter of comparing the embarked and disembarked numbers. Then using similar methods, we also get mortality rate and through that the mortality percentage.

Then, using the ggplot function in Rstudio several graphs were created which are included in the Github folder. These show slave voyages by decade, the top ten destination and origins for voyages and the average mortality rate by decade.

Analysis

The first graph (Bilag 1¹¹) created presents the number of recorded slave voyages per decade within the selected dataset from the SlaveVoyages archive between 1750 and 1800. The x-axis represents decades, while the y-axis shows the number of voyages recorded in the archive. The graph visualizes the development of the transatlantic slave trade across the selected period and illustrates how the frequency of voyage changed over the course of decades.

The graph demonstrates a clear increase in the number of recorded voyages from the mid-eighteenth century onward. The frequency of voyages rises steadily through the latter half of the eighteenth century and appears to peak toward the late 1700s before declining near the

¹¹ Bilag 1 in Appendix

turn of the century. This suggests that the slave trade intensified significantly during this period, which accurately reflects the expansion of Atlantic plantation economies and the increasing demand for slave labour in the Americas. The upward trend visible in the graph likewise reflects the growing scale and organization of the slave trade during the eighteenth century. The European colonial systems expanded and thus the trade increasingly became systemized and integrated commercially across the western world. The high concentration of voyages during the late eighteenth century corresponds with the historical peak of the transatlantic slave trade.

However, the graph should not be interpreted as a complete representation of all voyages that occurred during this period. The graph only shows the voyages recorded within the SlaveVoyages database. As discussed earlier the archive is based on fragmented historical documentation, meaning many voyages are missing from the picture. As such it illustrates patterns within the reconstructed dataset rather than a flawless historical record.

Overall, the graph shows that there was a substantial increase in recorded slave voyages between 1750 and 1800. Through this we can see how digital archives can be used to identify patterns in the transatlantic slave trade.

The second graph (Bilag 2¹²) created shows the ten most common recorded regions where a voyage began within the dataset. The x-axis represents the number of recorded voyages whilst the y-axis lists the regions identified as points of origin. This visualization provides an overview of which region were most connected to the slave trade during the chosen timeframe

One of the clearest patterns shown in the graph is the dominance of England and Wales, which appear far more frequently than any other region. This reflects the central role Britain played in the transatlantic slave trade during the late eighteenth century, particularly after the rapid expansion of British commerce and plantation economies. In the Caribbean and North America British merchants and ports became deeply integrated into the trade network.

Other regions shown such as France, the Netherlands and Brazil also appear prominently. This shows how trade functioned as an interconnected system involving multiple European colonial powers and territories. The presence of Brazilian regions such as Bahia and Pernambuco is significant as Brazil became one of the most prominent destinations for enslaved Africans during the late eighteenth century.

At the same time, the graph also demonstrates one of the limitations of the archive. The category “None,” appears in the graph due to incomplete historical records. This highlights an important methodological issue within the archived research. Being that it can only show the information that survived in the documentation, thus making missing or ambiguous records visible as gaps within the data itself. Additionally, the graph simplifies historical complexity.

¹² Bilag 2 in Appendix

Large regions are grouped together into broad categories, which vastly reduces local detail. Individual ports and changing political boundaries are largely done away with in favour of clearer visualization.

In the third graph (Bilag 3¹³) created presents the most common destinations for the slave ships. That is to say where they would disembark the slaves to be sold into labour. As with the origin graph there are some oddities to be found. Particularly in the category “0”. This represents incomplete shipping manifests that failed to specify where exactly the slaves were brought to. Common enough to make it the third highest result.

The most common destinations depicted in the graph add up with what we’ve already discussed. As the Caribbean became highly dependant on its plantation economy it needed a surplus of slave labour beyond what any other region did. Brazil coming in second is not surprising either. As discussed with former graphs there is the issue present here that multiple varied regions are compiled into larger denominators. Whilst the data isn’t wrong it fails to encompass the intricacies of the slave trade.

The fourth (Bilag 4¹⁴) and final graph created depicts the average mortality rate per decade. The enslaved people aboard the slave ships were trapped in inhumane conditions on a voyage across the great ocean and most often this lead to loss of life. At the start of the 1700s the percentage is at its peak with well over 16 percent of the enslaved dying during each voyage, however this number steadily declines over the century and curiously in the data presented falls to an alltime low between 1775 and 1800, where it steadily rises again.

This decline in lethality does not come from kindness or better quarters, rather as was previously mentioned the ships themselves were evolving.

From these graphs we have determined that a quantitative analysis of the fragmented historical sources is quite possible, but does leave room for interpretation and indeed improvement.

Discussion

The graphs produced via Rstudio are serviceable. They allow one to chart general trends in the development and growth of the transatlantic slave trade and the data presented follows what is known and documented about the development of the trade in the late eighteenth century. However, it is increasingly noticeable that dealing with something so complicated and massive as the transatlantic slave trade in a generalizing quantitative way leaves room for gaps. Incomplete historical documentation as well as missing documentation presents make way for uncertainty and incomplete data. These are the realities of reborn digital sources as

¹³ Bilag 3 in Appendix

¹⁴ Bilag 4 in Appendix

they are reliant on the quality of the historical source which, being a physical source, is subject to degradation, wear and tear. Though the knowledge we have certainly on the digital archive will not face similar troubles and has the potential to last if the internet remains.

There are limitations to the quantitative approach taken in this project. Whilst we can see the larger trends and developments of the largest influences on the transatlantic slave trade, there is an inherent simplification in it. Large regions are bundled into one, disregarding their intricacy as a society. We also greatly risk the aforementioned thingification where the weight of human suffering becomes little more than numbers and cargo. Once more repeating the dehumanization of the enslaved Africans and their individuality when we treat them as units of measurement rather than the living breathing people that they were. Scholars of slavery have long acknowledged the methodological difficulty on the reliance of archives fundamentally marked by profound violence, calculating mass mortality unavoidably depends on statistical data generated by the enslavers themselves in the attempts to control and discipline the enslaved.¹⁵ The call made by Jessica Marie Johnson for a “Black Digital Practice” has resulted in some digital humanists attempting to reconcile these issues through projects such as the Early Caribbean Digital Archive, Mapping Marronage, (Un)silencing Slavery and more. These works seek to use digital methods to illuminate the life practices of those enslaved so that we may understand them as more than mere cargo.¹⁶

Conclusion

SlaveVoyages.org demonstrates the significant potential of digital archives for historical research and quantitative analysis. Through its structured and accessible datasets, it becomes possible to analyze large-scale developments both through the decades and geographically within the transatlantic slave trade.

This project has shown that the database allows historians to chart broader developments in the expansion of the slave trade, despite gaps and inconsistencies within the surviving historical documentation. By using digital methods and quantitative analysis in RStudio, it was possible to identify patterns in voyage frequency and mortality rates across the selected period between 1750 and 1800.

At the same time, the project has also demonstrated the limitations inherent in digital archival research. The dataset is a reconstruction built upon fragmented historical records. This means that missing voyages, incomplete manifests and simplified classifications will shape the results of the analysis. The process of transforming human suffering into structured data

¹⁵ Landels, Tye, Isabel Bradley, Kelsey Desir, Grant Glass, Jane Harwell, Anya Lewis Meeks, and Charlotte Sussman. “Died a small boy”: Re-Centering the Human in Geospatial Data from the Middle Passage. P. 2

¹⁶ Landels, Tye, Isabel Bradley, Kelsey Desir, Grant Glass, Jane Harwell, Anya Lewis Meeks, and Charlotte Sussman. “Died a small boy”: Re-Centering the Human in Geospatial Data from the Middle Passage. P. 3

raises ethical and historiographical questions. As Jessica Marie Johnson argues, the reduction of enslaved people into numerical categories risks repeating the dehumanizing violence of the slave trade itself.

The methods of digital archives provide historians with powerful new analytical tools, but they also require critical thinking and the application of logic. Quantitative analysis makes it possible to identify patterns on a macro historical level while risking the loss of individual human experience and the historical nuance it brings. Those who work with digital archives must remain aware not only of what datasets reveal, but also of what they do not.

Ultimately, projects such as SlaveVoyages.org illustrate how digital methods are increasingly becoming an essential part of historical research. When approached critically digital archives allow historians to work with historical sources at a scale and level of accessibility not seen before in centuries of scholarship.

Statement on the use of Artificial Intelligence

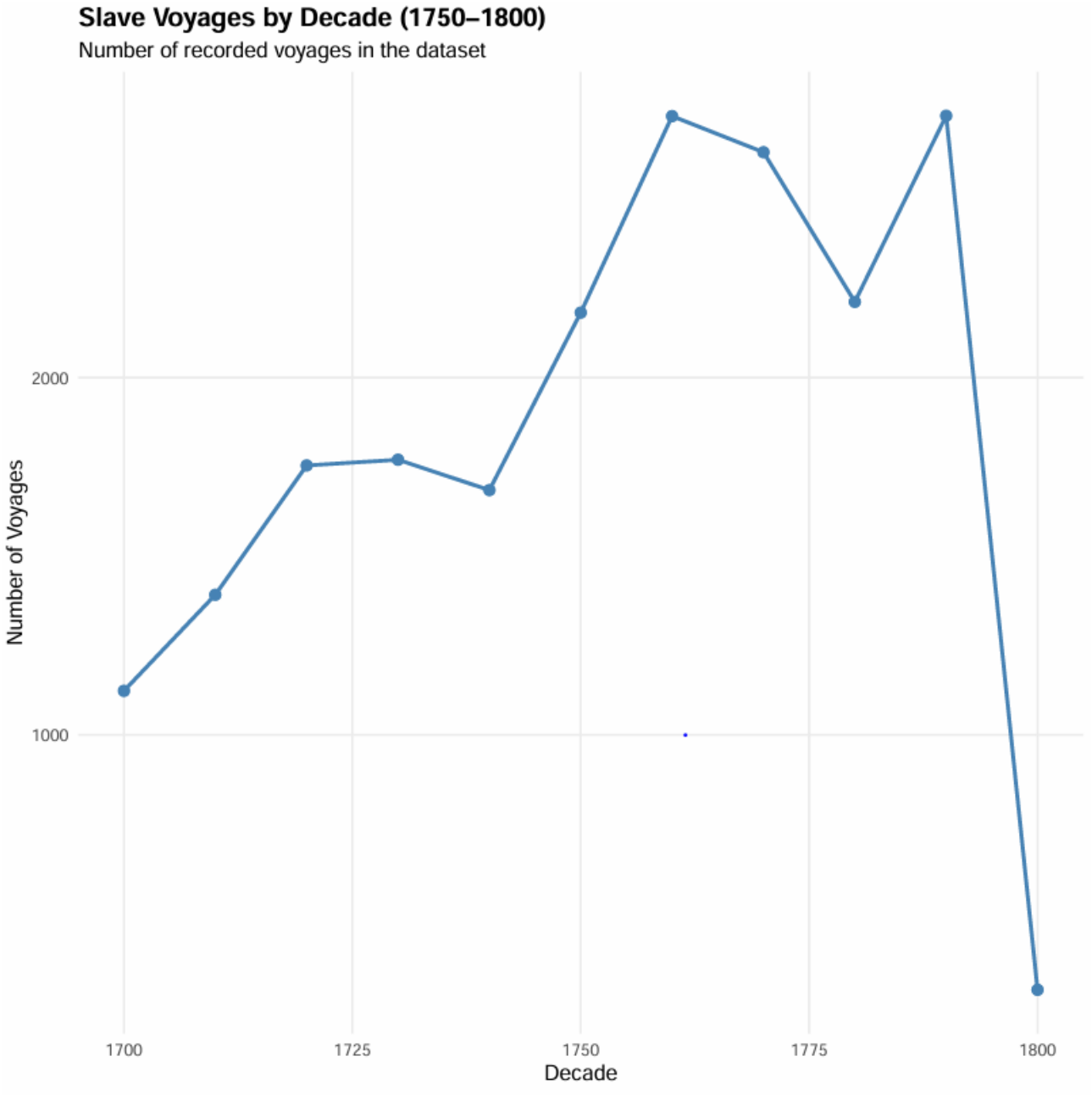
Artificial intelligence tools (ChatGPT by OpenAI, Notebook LM by Google) were used as supplementary support during the development of this project. AI assistance was primarily used for troubleshooting RStudio coding issues as well as improving the clarity and structure of academic English.

The AI tool was not used to generate the dataset from SlaveVoyages or to perform the historical analysis independently. Everything stated by the AI went under revision and evaluation by the author. All data selection, visualization, historical interpretation and written segments were created by the author with corrections made following by the AI.

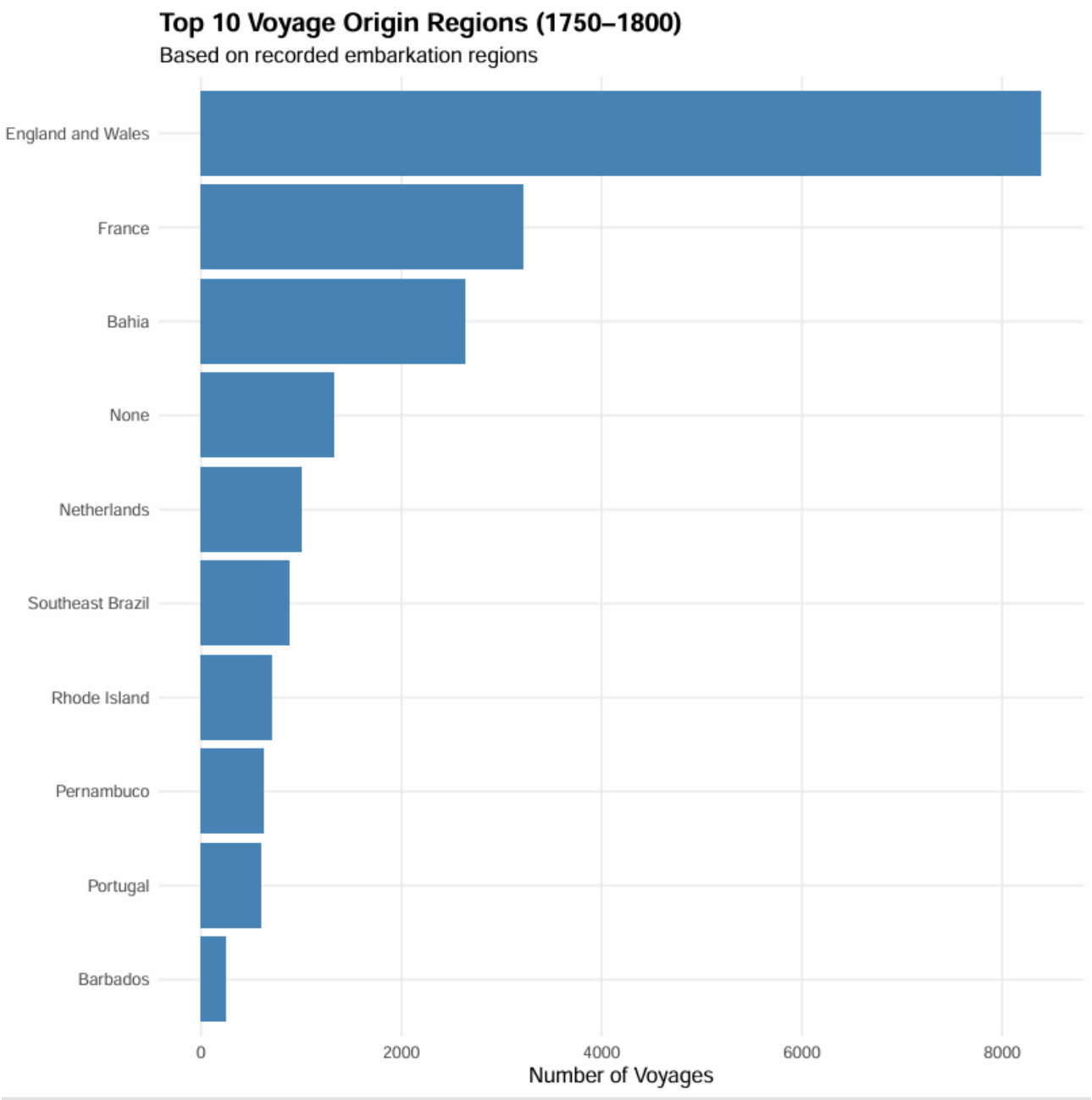
The use of AI functioned primarily as a digital support and feedback tool during the research and writing process.

Appendix

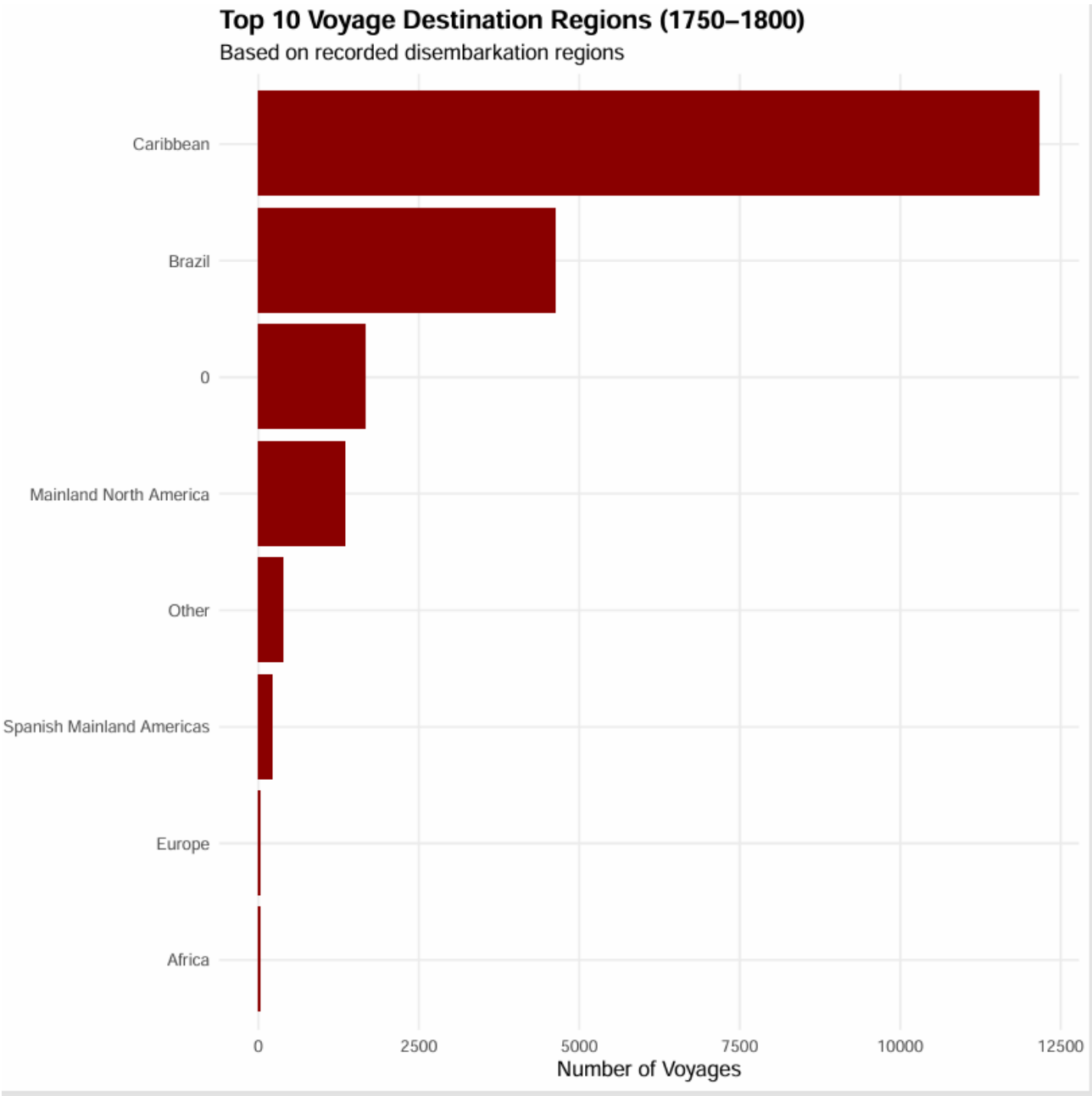
Bilag 1



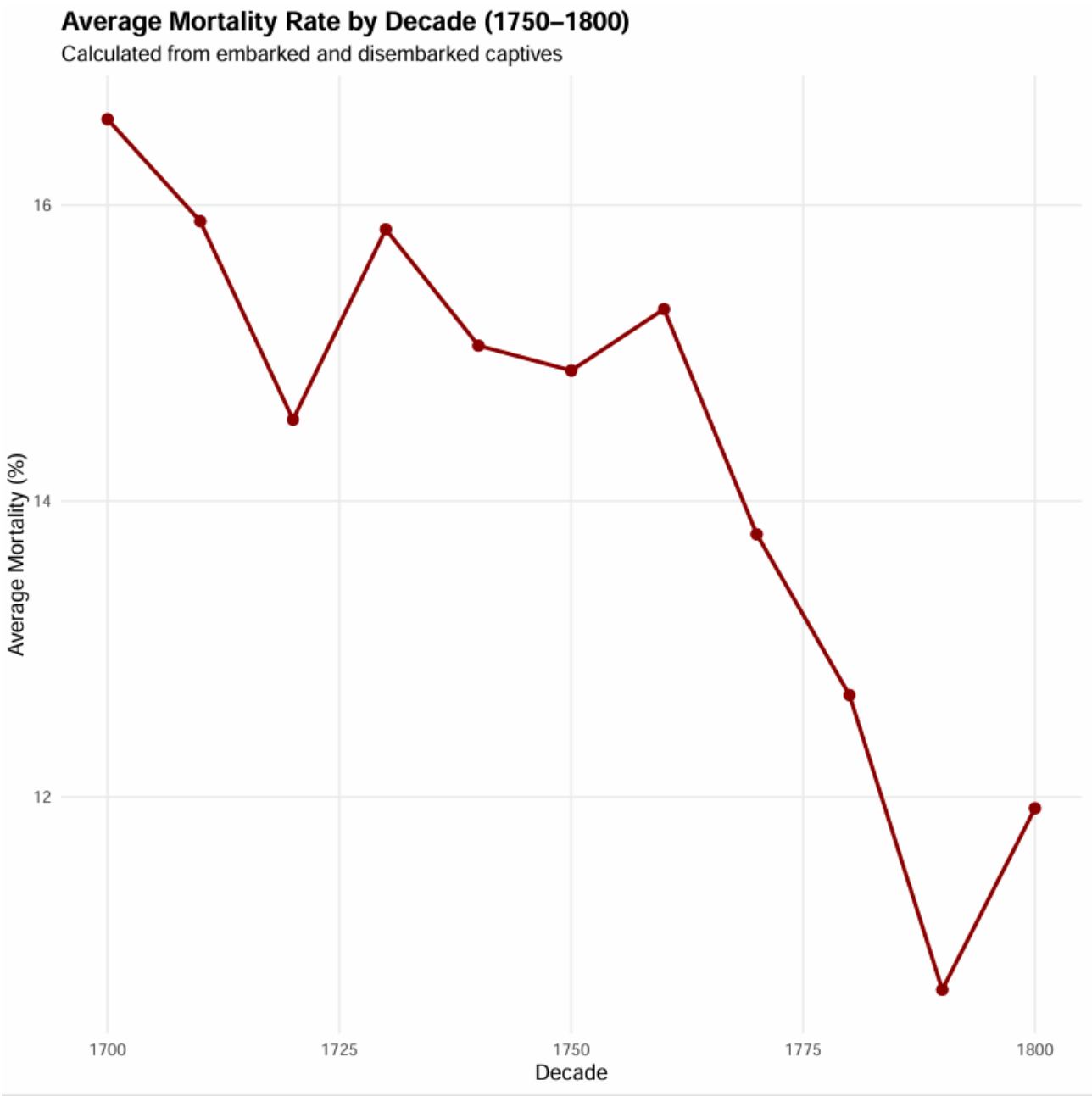
Bilag 2



Bilag 3



Bilag 4



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